# Fiber-Compatibility and Composite Suppression in Erdős–Straus Covering Families

Derek Blair
derekjblair@gmail.com

October 2026

## 1 Abstract

We report five formalized components of a program studying residue-covering families attached to the Erdős–Straus equation $4/n = 1/x + 1/y + 1/z$, all [PROVED] in Lean 4.34.1 with zero sorrys and standard-only axioms.

(1) The *Fiber-Compatibility Theorem*: for $\pi(P) \bmod p^e$ with Hensel digit sets $S_r$ over each base class $r \bmod p^{e-1}$, inverse-closure holds iff $S_{r^*} = A_r(S_r)$ on every inverse pair, where $A_r(u) = (-u(r^*)^2 - kr^*) \bmod p$ is the forced Hensel digit map. The $A_{r^*} \circ A_r = \mathrm{id}$ identity, verified computationally on all 10 prime powers in the discovery round, is now machine-checked. (2) The composite lift to $m = nD$ with $D \mid n$: the same iff, with the $D \mid n$ lemma killing the $n^2uu^*$ term the only new ingredient. The naive component-wise reduction is FALSE (Correlation Break Theorem [COMPUTED]): 12/44 surveyed composite moduli, and minimal 2-element witnesses at $36, 72, 100, 225$, project to inverse-closed sets on every prime-power component yet fail closure mod $m$ — the obstruction is a *correlation hole* (e.g. the $(2 \bmod 3, 0 \bmod 7)$ hole at $m = 441$). (3) Composite suppression theory: the suppression coefficient $\kappa(m \to m', r)$ and persistence rate $\Pi$, demands-descend (T1), the capability transfer law (T2), the no-lift theorems (T3), and the tower-refinement operator theory (T4–T8), including the $\Delta$-gating of L1 proliferation and the impossibility of predicting $\kappa$ from coarse data. (4) The realizable thinned-tiling catalogue on fibers $\{6, 11\}$: 16/16 shape×fiber verdicts BLOCKED as `decide`, live-union bounds 6/11 and 7/11, and the uniform no-tiling theorem closing every $m \geq 2$. (5) Branch-2 degeneracy for the Erdős–Straus splitting form: the "cubic" branch-2 identity collapses to a thin Diophantine condition ($M \mid r$, $4x = n + m$, integrality of $p, q$), the divisor-closed family theorem (92 targets as one infinite family $s \equiv 73 \pmod{264}$), and an explicit formal *refutation* of the naive mod-8 conclusion "$m \equiv 7 \pmod 8$".

Per the program's honesty discipline, mechanically decided results (the 16 BLOCKED verdicts, survey aggregates) are labeled [COMPUTED] and distinguished from deductive proofs; one stated conjecture is reported as refuted, not proved.

## 2 Introduction

The Erdős–Straus conjecture asks for every $n \geq 2$ a decomposition

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \qquad x, y, z \in \mathbb{Z}_{>0}. \tag{1}$$

Our program attacks (1) through residue-covering families: branch-1 splitting identities organized by a divisor-closed search at modulus $M = 2042040$, the degenerate branch-2 form, thinned multi-quad tilings of the $\kappa$-point fibers, and the modular-sieve survivor sets whose projections

$\pi(P)$ must be *inverse-closed* for the covering machinery to lift through prime powers and composite moduli. This preprint reports the five components of that machinery that have reached machine-checked status.

**Inverse-closure and fiber-compatibility.** For a finite $P \subset \mathbb{Z}$ and $m = p^e$, let $\pi$ be projection mod $m$. Write every unit lift uniquely as $x = r + p^{e-1}u$ and define the *Hensel digit set* over each base class $r$ mod $p^{e-1}$,

$$S_r = \{\, u \in \{0, \ldots, p-1\} : r + p^{e-1}u \in \pi(P) \,\}. \tag{2}$$

When is $\pi(P)$ inverse-closed, i.e. $x^{-1} \bmod m \in \pi(P)$ for every unit $x \in \pi(P)$? Round 185 discovered the sharp answer: closure holds iff digit-compatibility $S_{r^*} = A_r(S_r)$ holds on every inverse pair, where $A_r$ is the forced Hensel digit map derived from Bézout expansion of the inverse [1]. The theorem explains three breaks in the chain data (at 169, 289, 343) as per-prime selection-history effects, and proves that saturation is sufficient but not necessary ($m = 8$ is closed with unsaturated fibers) [1]. Rounds 190/194 machine-checked the kernel and the full lift-set packaging [2, 3].

**Composites and correlation holes.** Round 192 lifted the theorem to general composite modulus via CRT, and found that the naive reduction to prime-power components is false: a set can project to closed sets on every component yet fail closure mod $m$, because inversion couples components and the *joint* digit set can hold a correlation hole even when each component projection is compatible [4]. Rounds 198/206 machine-checked the composite kernel [5, 6]. Rounds 205/207 then invented the suppression theory governing how correlation holes behave under modulus refinement: the coefficient $\kappa(m \to m', r)$, the persistence rate $\Pi$, the descend/lift laws T1–T3, and the tower-refinement operator theory T4–T8 with its $\Delta$-gating of L1 proliferation [9, 10]; Rounds 210/214 machine-checked the finite-set kernels and recorded the survey data as decidable facts [11, 12].

**Tilings on the last two fibers and branch 2.** Round 193 decided the realizable thinned-tiling catalogue on the two un-thinned fibers 6 and 11 — all 16 shape×fiber verdicts BLOCKED — and closed the no-tiling argument on every multiplicity via a live-union bound, extending the program's 10-fiber theorem to all 12 fibers [7]; Round 202 machine-checked the finite core and the uniform argument [8]. Round 209 proved the branch-2 degeneracy theorem (the "cubic" identity collapses to the linear relation $4x = n + m$ plus integrality), ran the divisor-closed $r$-search to completion (92/288 reachable, one infinite family), and Round 218 machine-checked it — and in doing so formally refuted the naive mod-8 conclusion of part (4) [13, 14].

Our contributions, all [PROVED] unless labeled:

1. The fiber-compatibility kernel for prime powers (Theorem 1) and its full lift-set packaging (`FiberCompatLift.fiber_compat_iff`).
2. The composite iff with the $D \mid n$ lemma, the (injective-only) projection theorem, and the Correlation Break [COMPUTED] (Theorem 2, Proposition 1).
3. Suppression kernels (T1, T2(a)), the $\kappa/\Pi$ calculus, the R-operator section T4–T8, and the survey data (Theorems 3–6).
4. The fiber-$\{6, 11\}$ catalogue as `decide` and the uniform no-tiling theorem (Theorem 7).
5. Branch-2 degeneracy (1)–(3),(5), the family identity, the $s = 49$ witness, and the refutation of the mod-8 claim (Theorems 8, 9; Refuted claim 4.5).

# 3 Methods

## 3.1 Definitions

**Fiber data (prime power).** Fix a prime $p$ and $e \geq 2$. For a finite $P \subset \mathbb{Z}$, $\pi : \mathbb{Z} \to \mathbb{Z}/p^e$ the projection, define $S_r$ by (2). For a unit base class $r \bmod p^{e-1}$ let $r^* = r^{-1} \bmod p^{e-1}$ and $k = (rr^* - 1)/p^{e-1} \in \mathbb{Z}$. The *forced Hensel digit map* is

$$A_r(u) \equiv -u(r^*)^2 - kr^* \pmod{p}, \tag{3}$$

an affine bijection of $\mathbb{Z}/p$ (slope $-(r^*)^2$ a unit) with $A_{r^*} \circ A_r = \mathrm{id}$ (proved, not assumed: R185 verified it computationally on all 10 prime powers; R190 machine-checked it as `A_map_invol`).

**Composite data.** Let $m = \prod p_i^{e_i}$ with distinct primes, $E = \{i : e_i \geq 2\}$, $n = \prod_{i \in E} p_i^{e_i - 1} \prod_{i \notin E} p_i$ (squarefree primes folded into the base), $D = \prod_{i \in E} p_i$ (squarefree), so $m = nD$ and

$$D \mid n, \tag{4}$$

since $p_i \mid p_i^{e_i - 1}$ for $i \in E$. Every $x \bmod m$ writes uniquely as $x = r + nu$ with $r \in \mathbb{Z}/n$, $u \in \mathbb{Z}/D$ (CRT). The composite digit set is

$$S_r = \{\, u \in \mathbb{Z}/D : r + nu \in \pi_m(P) \,\}, \tag{5}$$

and the forced map is (3) with $p$ replaced by $D$ and $k = (rr^* - 1)/n$ [4].

**Suppression data.** At nested composite moduli $m \mid m'$ (R192 form), a fiber pair $(\tilde{r}, \tilde{r}^*)$ at $m'$ *covers* $(r, r^*)$ at $m$ if $\{\tilde{r} \bmod n, \tilde{r}^* \bmod n\} = \{r, r^*\}$ (set-normalized). For a demands-holding, hole-capable pair $(r, r^*)$ at $m$ with covering set $C$ of demands-holding, hole-capable covers at $m'$, the *composite suppression coefficient* is

$$\kappa(m \to m', r) = \frac{|\{c \in C : \sigma(c) = 1\}|}{|C|}, \tag{6}$$

undefined if $C = \varnothing$; $\sigma(c) = 1$ means the cover's correlation hole stays suppressed. The *persistence rate* $\Pi(m \to m')$ is the fraction of suppressed-at-$m$ pairs (with $C \neq \varnothing$) having $\kappa = 1$ [9]. The tower-refinement operator $R_{m \to m'}$ packages the cover system, the per-prime affine digit law $u \equiv v + L\tilde{u} \pmod{p}$ (with $L = n'/n$), the *degeneracy set* $\Delta(m \to m') = \{p \in E : p \mid L\}$, and the growth signature [10].

**Tiling data.** On a fiber, the 11 $\kappa$-points are covered by 47 box instances $(S, \text{roots})$; each instance is RULED-OUT (0), OPEN (1), or ACHIEVABLE (2) by the $(c \bmod 5005, S)$ SAT decision procedure. A shape is BLOCKED iff some constituent instance is RULED-OUT; a *live* instance is ACHIEVABLE or OPEN. The catalogue shapes are the 7 minimal 3-covers and the unique 2-cover [7].

## 3.2 Formalization note

All five components are formalized in the `derekblair/math-proofs` GitHub repository, directory `Mathproofs/`, machine-checked on bakkie (the standing Lean machine) with `lake build` exit 0, zero sorrys, and axioms within [propext, Classical.choice, Quot.sound]:

- `Mathproofs/ESFiberCompat.lean` (R190/R194): `A_map_invol` (the involution, via a `linear_combination` certificate), `forced_digit` (the Bézout heart), `fiber_compat_pair_iff` (pair-level iff), `fiber_compat_saturated` (Cor. 1), `fiber_compat_card` (Cor. 2), `fiber_compat_partial` (Cor. 3, $m = 8$ by `decide`); the `FiberCompatLift` namespace adds the full $\pi(P)$ packaging `fiber_compat_iff` with `isUnit_proj`, `proj_inv`, `Kquot_spec`.

- Mathproofs/ESFiberCompatComposite.lean (R198/R206): forced_digit_dvd (the $D \mid n$ lemma), fiber_compat_pair_iffC, proj_inv_closed and proj_inv_closed_zmod (the ($\Rightarrow$) projection theorem for any commutative ring), forced_digit_of_memC, fiber_compat_iffC.

- Mathproofs/CompositeSuppression.lean (R210/R214): demands_descend (T1 kernel), margin_nontrivial_image, capability_descend (T2(a) kernel), the $\kappa/\Pi$ calculus (kappa_mem_Icc, kappa_eq_one_iff, kappa_eq_zero_iff, persistRate_mem_Icc), the survey theorems (towerKappa225_values, towerKappa225_not_constant, mean_kappa_aggregate, destruction_L_mod5, desaturation_refinement, levelchange_refinement, census_headlines), and the R-operator section (digitMap, degeneracy_lemma, const_digit_image, bijective_digit_nontrivial_iff, union_bijective_images_nontrivial, kappa_ne_one_of_desaturation, kappa_one_of_singleton, impossibility_witness1/2). The survey theorems record [COMPUTED] census data as decidable facts; the empirical mechanism counts behind them are cited, not re-proved, per the module's convention.

- Mathproofs/ESFiberTiling.lean (R202): the 16 shape verdicts as decide (shape0_fiber6_blocked through shape6_fiber11_blocked, cover2_fiber6_blocked, cover2_fiber11_blocked), liveUnion_bound_6, liveUnion_bound_11, unique_2cover, seven_minimal_3covers, and the uniform theorems no_tiling_fiber6, no_tiling_fiber11. The 94 verdict table is [COMPUTED] data, kernel-checked only as given tables.

- Mathproofs/ESBranchTwo.lean (R218): branch2_M_dvd_r, branch2_four_x_eq, branch2_collapse, branch2_m_le_s, branch2_divisibility_shape, branch2_ab_dvd_XSTEP, family_integrality, family_identity, branch2_witness_s49, and the refutation branch2_m_mod8_refuted (with its branch2_mod8_counterexample_hyps/identity lemmas).

Lean 4.34.1, Lake 5.0.0, Mathlib v4.34.1 (pinned commit d13f23b723b8a846827a245b89c10fc7d3f11612). The module headers and this paper cite each other by name and by the slug fiber-compatibility-composite-s per the program's cross-reference rule.

### 3.3 Honesty discipline and assumptions

[PROVED] means machine-checked in Lean with zero sorrys and the axioms above. [PROVED*] means a hand proof exists with a named formalization target outstanding. [COMPUTED] means verified by two independent code paths (or a kernel decide) but not deductively proved. [OPEN] means an explicit question with no answer. [REFUTED] means a stated claim proved false. Only [PROVED] results get proof-claiming language.

**Assumption 1** (Survivor-data lineage)**.** The residue data (parents196, the R67→R71→R74 chain, the 44 surveyed composite moduli, the 33 nested refinement pairs) and the empirical iff/census numbers drawn from the dossiers are supplied datums: computed under two independent code paths with the verification logs cited in the dossiers, but not re-derived in this preprint. Figures and tables that tabulate them are marked [COMPUTED].

**Assumption 2** (SAT decision procedure completeness)**.** The ($c$ mod 5005, $S$) procedure's RULED-OUT verdicts are sound on the complete search box $c \le 7507$, $h \le 2000$ (R55, cited): an instance is RULED-OUT only if every death-avoiding candidate lies inside the box with zero JSON witnesses. The 16 BLOCKED verdicts and the live-union bounds are decide theorems over the resulting verdict tables; the procedure's correctness itself is [COMPUTED]-external.

**Assumption 3** (Cited tiling premises)**.** The tiling criterion, coordinate+box+slice lemmas, the $h$-bound, the complementary-5-slice global impossibility (R184), the 10-fiber census and minimal 3-cover shapes (R188), and the un-thinned census (R180) are cited as premises, not re-proved. The Lean no-tiling theorem is end-to-end conditional on them.

# 4 Results

## 4.1 The Fiber-Compatibility Theorem [PROVED]

**Theorem 1** (Pair-level fiber-compatibility; PROVED as `fiber_compat_pair_iff`)**.** *Let $s, t, k \in \mathbb{Z}/p$ with $ts = 1$ and $s \neq 0$. For digit sets $D, D' \subset \mathbb{Z}/p$,*

$$\mathrm{PairClosed}(s, k, D, D') \wedge \mathrm{PairClosed}(t, k, D', D) \iff D' = \mathrm{image}(A_{\mathrm{map}}(s, k), D),$$

*where* $\mathrm{PairClosed}(s, k, D, D')$ *is the statement that every $u \in D$ has its forced digit $A_{\mathrm{map}}(s, k)(u) \in D'$.*

The ($\Rightarrow$) direction is the card argument: inclusion both ways via the involution `A_map_invol` ($A_{r^*} \circ A_r = \mathrm{id}$), equal cards via injectivity, closing with `Finset.eq_of_subset_of_card_le`. The involution itself, computationally verified on all 10 prime powers in R185, is promoted to [PROVED] here with a `linear_combination` certificate.

The Bézout heart is `forced_digit`: if $(r + p^{e-1}u)(s + p^{e-1}v) \equiv 1 \pmod{p^e}$ with $rs = 1 + p^{e-1}k$, then $v \equiv -us^2 - ks \pmod{p}$ — the inverse's digit is *forced* to be $A_r(u)$. The full $\pi(P)$ packaging is `FiberCompatLift.fiber_compat_iff`: $\pi(P)$ inverse-closed iff digit-compatibility holds at every unit base class (R194), promoting R185's [PROVED*] theorem to [PROVED] at the kernel level.

**Corollary 1** (Saturation sufficient; PROVED as `fiber_compat_saturated`)**.** *Full fibers ($|S_r| = p$ for all $r$) imply compatibility, since the image of* univ *under the surjection $A_r$ is* univ*.*

**Corollary 2** (Fiber-asymmetry break; PROVED as `fiber_compat_card`)**.** *Compatibility implies $|S_r| = |S_{r^*}|$: fiber-size asymmetry is a pure counting refutation of closure. Instantiated in the chain data at 289's inverse pair $\{3, 6\}$ (7 vs. 9 digits) [1].*

**Corollary 3** (Partial-but-closed exists; PROVED as `fiber_compat_partial_closed`)**.** *At $m = 8$: $S_1 = \{0\}$ (only 1 mod 8 present), $A_1(u) = -u \bmod 2$ fixes $\{0\}$, so closure holds with an unsaturated fiber (by `decide` on $\mathbb{Z}/2$). Hence "unsaturated $\Rightarrow$ broken" is false; digit-compatibility is the sharp condition.*

Figure 1 illustrates the sharpness on real chain data: at $169 = 13^2$, the inverse pair $(8, 5)$ has $S_8 = [0, 1, 4, 9, 11]$ and $S_5 = [0, 1, 3, 4, 11]$, while compatibility demands $A_8(S_8) = [2, 7, 9, 11, 12]$ — equal sizes $(5, 5)$, a genuine digit mismatch, not asymmetry [COMPUTED] [1]. The violator $(5, 34)$: $34 = 8 + 13 \cdot 2$ needs digit $2 = A_5(0)$, but $2 \notin S_8$ (removed by R74, fiber 8 going $10 \to 5$).

The empirical iff held on all 10 probed prime powers $(8, 9, 25, 27, 49, 121, 125$ closed; $169, 289, 343$ broken) on two independent code paths [COMPUTED] [1]; the mechanism per prime is a function of the chain's residue/selection history at $p$, not of $(p, e)$ — e.g. 169 breaks despite $196 > \varphi(169) = 156$ [1].

## 4.2 The composite lift and the Correlation Break

**Theorem 2** (Composite fiber-compatibility; PROVED as `fiber_compat_iffC`)**.** *For $m = nD$ with $D \mid n$, a finite set $P$ of units mod $m$ is inverse-closed iff $S_{r^{-1}} = \mathrm{image}(A_r, S_r)$ at every unit base class $r$.*

The proof is R185's proof verbatim once the $n^2uu^*$ term vanishes mod $D$ — which is exactly (4). That vanishing is the only new mathematics, formalized as `forced_digit_dvd`: with $rs = 1 + nk$ and $(r + nu)(s + nv) = 1 + mt$, where $nD \mid m$ and $D \mid n$, we get $D \mid v + us^2 + ks$. The naive ($\Rightarrow$) direction also survives in full generality: `proj_inv_closed` shows a ring homomorphism sends inverse-closed sets to inverse-closed sets (stated without a global inverse, for any commutative ring) — so composite-closed implies every prime-power component closed [PROVED*].

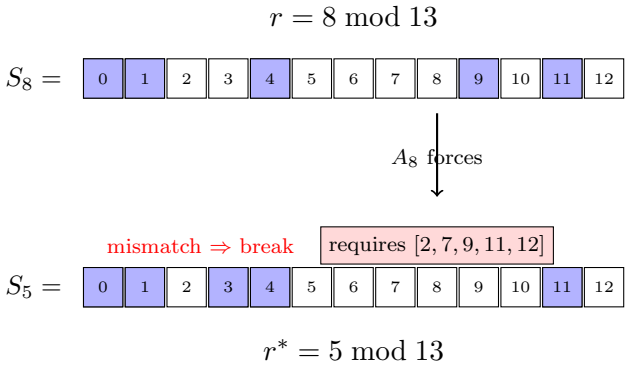



Figure 1: Digit-compatibility failure at $169 = 13^2$ [COMPUTED, from R185]: the inverse pair $(8, 5)$ has equal-size digit sets $S_8 = [0, 1, 4, 9, 11]$, $S_5 = [0, 1, 3, 4, 11]$, but $A_8(S_8) = [2, 7, 9, 11, 12] \neq S_5$. The violator $(5, 34)$: $34 = 8 + 13 \cdot 2$ needs $2 = A_5(0) \notin S_8$. This is a genuine digit mismatch, not fiber-size asymmetry.

**Proposition 1** (Correlation Break Theorem; COMPUTED)**.** *The converse is false. Component-wise R185-compatibility on every prime-power factor does* not *imply inverse-closure mod $m$.*

Breaks occur in the real chain data on 12/44 surveyed composite moduli (441, 1089, 1125, 1225, 1323, 3025, 32 and minimally: exhaustive pair search finds 2-element correlation witnesses at 36 (4), 72 (24), 100 (18), 225 (22) [COMPUTED, $\times 2$ independent paths] [4]. Minimal hand witness at $225 = 9 \cdot 25$: $P = \{101, 149\}$; $\pi_9(P) = \{2, 5\}$ closed ($2^{-1} = 5 \bmod 9$), $\pi_{25}(P) = \{1, 24\}$ self-inverse, yet $101^{-1} = 176 \bmod 225 \notin P$. Digit anatomy ($n = D = 15$): $S_{11} = \{6\}$, $S_{14} = \{9\}$ (self-inverse bases), $A_{11}(\{6\}) = \{11\} \neq \{6\}$, $A_{14}(\{9\}) = \{4\} \neq \{9\}$ — product-incompatible with compatible projections. Real-data anatomy at $441 = 9 \cdot 49$ ($n = D = 21$): violator $(340, 310)$; $r = 4$, $r^* = 16$; $S_4$ full (21/21 digits), $S_{16}$ missing digit 14 = (2 mod 3, 0 mod 7); $A_4(S_4)$ demands $14 \in S_{16}$ $\times$. Per-component: $340 \bmod 9 = 7$ with inverse 4 present, $340 \bmod 49 = 46$ with inverse 16 present — each component's fiber contains its digit, but the *combination* never co-occurs: the correlation hole, in the wild [4].

### 4.3 Composite suppression theory [PROVED kernels; survey data COMPUTED]

Round 205 invented suppression under CRT modulus refinement; Round 207 built the tower-refinement operator theory behind its destruction mechanisms. We state the theorems with their Lean status; proofs are the dossier hand proofs, the finite-set kernels being machine-checked.

**Theorem 3** (T1, demands descend; kernel PROVED as `demands_descend`)**.** *Fine demands-holding implies coarse demands-holding.*

*Proof sketch [PROVED*]:* write $L = n'/n$; for $x$ in cover $\tilde{r} = r + nv$ the coarse digit $u \equiv v + L\tilde{u} \pmod{D}$ is the fixed affine image of the fine digit; applying the same affine map per cover and unioning over covers gives the coarse demand equalities. The kernel (image commutes with indexed unions) is `demands_descend`; 42/42 fine-demands covers confirmed [COMPUTED]$\times 2$ [9].

**Theorem 4** (T2, capability transfer; (a) kernel PROVED as `capability_descend`, (b) COMPUTED)**.** *(a)* Downward lemma: *a hole-capable cover whose non-singleton margins lie over two distinct coarse $E$-primes, with $L$ coprime to both, forces the coarse pair hole-capable. (b)* Upward is not a law: *of* 20 *coarse hole-capable pairs,* 18 *keep every cover hole-capable and* 2 *lose all capability, both via total fine-demands breaking (*$1225 \to 6125$ $(11, 16)$ *and* $3025 \to 15125$ $(6, 46)$*, both* $L \equiv 0 \pmod 5$ *exponent-growth).*

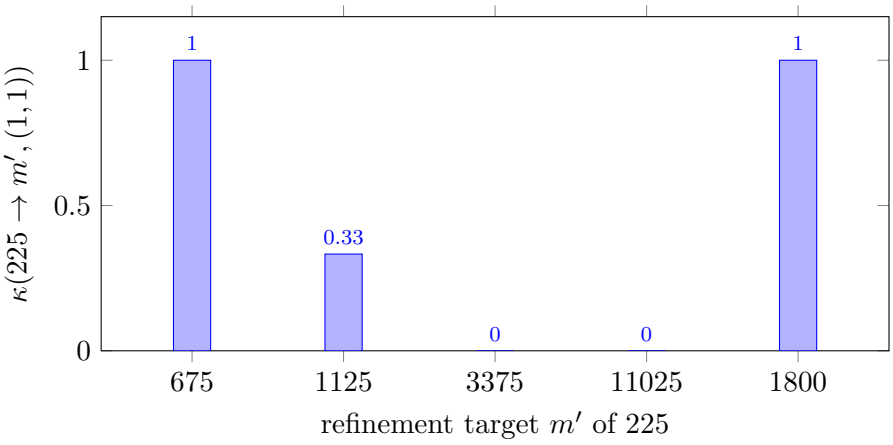



Figure 2: Data-specificity of suppression [COMPUTED, from R205]: the same coarse pair $(225, (1,1))$ — identical margins, $\sigma$, saturation, Hensel maps — has $\kappa = 1.0, 1/3, 0, 0, 1.0$ across five refinement towers. $\kappa$ is not a function of coarse data.

The kernel: an injective digit map sends non-singleton fine margins to non-singleton coarse margins (`margin_nontrivial_image`), and two disjoint non-singleton fine margins descend to two disjoint non-singleton coarse margins (`capability_descend`) [11].

**Theorem 5** (T3, suppression does not lift; PROVED* by witness)**.** *(a)* Desaturation: *T5 saturation forcing does not transfer upward — 225's saturated pair* $(1,1)$ *($\sigma = 1$) refines into a realized L2 hole at* 1125 *(cover* $(16, 61)$*, margins* $[3, 5]$*, $\sigma = 0$). (b)* Data-specificity: *the same pair* $(225, (1,1))$ *has* $\kappa = 1.0/0.333/0.0/0.0/1.0$ *across the five towers* $\to 675/ \to 1125/ \to 3375/ \to 11025/ \to 1800$ *(Figure 2): lifting is tower-specific, a fact about the refinement, never about $m$. Aggregate:* 10 *suppressed coarse pairs with hole-capable covers, only* 2 *persist fully, mean* $\kappa = 7/30 \approx 0.233$*. (c)* Level change: *441's L2 pair* $(4, 16)$ *refines into a fine-L1 cover* $(25, 58)$ *at* 1323.

The five $\kappa$-values, the desaturation refinement, the level change, and the census headlines are recorded as survey theorems (`towerKappa225_values`, `towerKappa225_not_constant`, `mean_kappa_aggregate`, `desaturation_refinement`, `levelchange_refinement`, `census_headlines`); the witnesses and aggregates stay [COMPUTED]×2 [9, 11].

**Theorem 6** (T4–T8, the R-operator; kernels PROVED, witnesses survey)**.** *(T4) The per-prime digit map* $\varphi_v(\tilde{u}) = v + L\tilde{u} \bmod p$ *is a bijection of* $\mathbb{Z}/p$ *iff* $p \notin \Delta$*, and the constant map $v$ iff $p \in \Delta$ (`degeneracy_lemma`). (T5) L1 proliferation is $\Delta$-gated: $p \in \Delta$ allows* unilateral *breaking (constant digit map collapses any fine margin to a singleton, `const_digit_image`); $\Delta = \varnothing$ forces cross-cover compensation, and with a single active cover fine demands iff coarse demands (`bijective_digit_nontrivial_iff`, `union_bijective_images_nontrivial`). (T6) $\kappa = 1$ implies no hole-capable cover desaturates (`kappa_ne_one_of_desaturation`); total L1 proliferation makes $C = \varnothing$ — capability destruction. (T7) $\Delta = \varnothing$ + single active cover + fine T5-saturation implies $\kappa = 1$ (`kappa_one_of_singleton`). (T8) No function of (coarse margins, $\sigma$, T5 status, $\Delta$, $L$, growth pattern) predicts $\kappa$:* $225 \to 675$ *($\kappa = 1.0$) vs* $1125 \to 3375$ *($\kappa = 0.0$) agree in all six features, as do* $225 \to 1125$ *($\kappa = 0.333$) vs* $675 \to 3375$ *($\kappa = 0.0$) (`impossibility_witness1/2`); the obstruction is fine-level data — coarse saturation does not shard into fine saturation (675's covers saturated* 15/15 *vs 3375's starved* $\{11, 9, 10\}$*).*

Empirical mechanism counts: the affine digit law 3332/3332 point-checks, the margin-union identity 48/48, and the gating census (22/22 census + 11/11 sweep demands-breaking covers at $\Delta \neq \varnothing$ steps, 0 at any $\Delta = \varnothing$ step) are [COMPUTED]×2, cited as the survey content behind the kernels; $\Delta$ is necessary for L1 proliferation, not sufficient (degenerate steps with zero breaking exist, e.g. $225 \to 675$ 0/2) [10, 12].

Table 1: The realizable thinned-tiling catalogue on fibers $\{6, 11\}$ [COMPUTED, from R193]: 7 minimal 3-cover shapes $\times$ 2 fibers plus the 2-cover, all BLOCKED. RO = RULED-OUT, A = ACHIEVABLE. Each verdict is a `decide` theorem in `ESFiberTiling.lean`.

| shape | fiber 6 | fiber 11 |
|---|---|---|
| #0 $(\{5\},0)+(\{11\},3)+(\{11\},6)$ | RO,RO,RO $\to$ BLOCKED | RO,A,RO $\to$ BLOCKED |
| #1 $(\{5\},0)+(\{11\},3)+(\{5,11\},(2,6))$ | RO,RO,RO $\to$ BLOCKED | RO,A,RO $\to$ BLOCKED |
| #2 $(\{5\},0)+(\{11\},6)+(\{5,11\},(2,3))$ | RO,RO,RO $\to$ BLOCKED | RO,RO,RO $\to$ BLOCKED |
| #3 $(\{5\},0)+(\{5,11\},(2,3))+(\{5,11\},(2,6))$ | RO,RO,RO $\to$ BLOCKED | RO,RO,RO $\to$ BLOCKED |
| #4 7-tiling $(\{7\},0)+(\{7\},1)+(\{7\},5)$ | RO,RO,RO $\to$ BLOCKED | RO,RO,RO $\to$ BLOCKED |
| #5 11-tiling $(\{11\},3)+(\{11\},6)+(\{11\},8)$ | RO,RO,RO $\to$ BLOCKED | A,RO,RO $\to$ BLOCKED |
| #6 $(\{11\},3)+(\{11\},6)+(\{5,11\},(0,8))$ | RO,RO,RO $\to$ BLOCKED | A,RO,RO $\to$ BLOCKED |
| 2-cover $(\{5\},0)+(\{5\},2)$ | RO,RO $\to$ BLOCKED | RO,RO $\to$ BLOCKED |

Table 2: Live-union bounds [PROVED as `decide`]: even granting every OPEN instance realizable, the achievable class coverage on fibers 6 and 11 stays below 11, closing tilings of every size at once.

| fiber | live instances | classes covered | bound |
|---|---|---|---|
| 6 | 7 | $\leq 6$ | `liveUnion_bound_6` |
| 11 | 5 | $\leq 7$ | `liveUnion_bound_11` |

### 4.4 The fiber-$\{6, 11\}$ tiling catalogue [PROVED as `decide`; SAT layer COMPUTED]

**Theorem 7** (No thinned multi-quad tiling on fibers 6 or 11; PROVED as `no_tiling_fiber6`, `no_tiling_fiber11`)**.** *No family of live boxes covers all* 11 $\kappa$*-points on fiber* 6 *or fiber* 11*; hence no thinned multi-quad tiling exists on either fiber for any* $k \geq 2$*.*

The uniform argument: every tiling box is *live* (a RULED-OUT instance cannot be realized, by the contrapositive of the decision rule under Assumption 2), so any cover's union lies inside the live union; but the live union covers at most 6 classes on fiber 6 and 7 on fiber 11 — `liveUnion_bound_6`, `liveUnion_bound_11` (both by `decide`; Table 2). The $m = 2$ leg is the unique 2-cover (`unique_2cover`, BLOCKED) and the $m = 3$ leg is the 7 minimal 3-covers (`seven_minimal_3covers`, all BLOCKED); Table 1 gives all 16 shape$\times$fiber verdicts as `decide` theorems. This extends the program's 10-fiber theorem to all 12 fibers; the 10 OPEN micro-instances (R193 §3) are individually undecided but provably tiling-irrelevant via the live-union bound. The 2 ACHIEVABLE instances — fiber 6: $(\{5, 7\}, \text{roots}\,(0, 1))$ covering 1 class; fiber 11: $(\{11\}, \text{root}\,3)$ covering 5 classes — never complete a shape [7, 8].

### 4.5 Branch-2 degeneracy, the family identity, and the mod-8 refutation

We work in the cubic framework at modulus $M = 4 \cdot \text{XSTEP}$ with the branch-2 ansatz $4/(Mt + s) = 1/x + 1/(nAC) + 1/(BC)$ for all $t \geq 0$, $x = \text{XSTEP}\,t + x_0$, $x_0 = (s + m)/4$, $m = 4c - 1$, $n = Mt + s$, $B = rt + w$, $C = pt + q$, $A, r, p, q$ positive integers.

**Theorem 8** (Branch-2 degeneracy; PROVED)**.** *(1)* `branch2_M_dvd_r`*:* $w = rs/M \in \mathbb{Z}$ *and* $\gcd(s, M) = 1$ *force* $M \mid r$ *— so every branch-2 search with* $r < M$ *(R67:* $r \leq 800 < M = 120120$*; R71:* $r \leq 5000 < M = 2042040$*) was provably vacuous. (2)* `branch2_four_x_eq`*,* `branch2_collapse`*:* $B = r_1 n$ *with* $r = Mr_1$*, and the identity collapses to* $4x = n + m$ *(true by definition of* $x_0$*) with* $(r_1 + A)x = mAr_1C$ *— the only nontrivial content is integrality of* $p = (r_1 + A)\text{XSTEP}/(mAr_1)$ *and* $q = (r_1 + A)x_0/(mAr_1)$*. Branch 2 is not a cubic family; it is a*

*thin Diophantine condition. (3)* `branch2_m_le_s`*:* $m \le s$*. (5)* `branch2_divisibility_shape`*,* `branch2_ab_dvd_XSTEP`*:* $Ar_1 \mid (r_1 + A)\text{XSTEP}$*, and with* $A = g_1 a$*,* $r_1 = g_1 b$*,* $\gcd(a, b) = 1$*:* $ab \mid \text{XSTEP}$*.*

**Theorem 9** (Divisor-closed family; PROVED as `family_integrality`, `family_identity`)**.** *The* 92 *newly reachable targets are exactly* $s \equiv 73 \pmod{264}$*, all sharing* $(c, m, A, r) = (4, 15, 1, 46410)$*: for every* $k$ *(*$s = 264k + 73$*) and every* $t$*,*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{nC} + \frac{1}{BC}, \quad \begin{array}{ll} n = 2042040t + s, & x = 510510t + (s+15)/4, \\ B = 46410t + (s+15)/44, & C = 1531530t + (3s+1)/4, \end{array} \tag{7}$$

*with* $44 \mid s+15$ *and* $4 \mid 3s+1$ *for* $s \equiv 73 \pmod{264}$*. Coverage rises* $110/398 \to 202/398$ *(*$50.8\%$*) [COMPUTED]; each of the* 92 *identities independently verified by polynomial-numerator plus exact* `Fraction` *checks [COMPUTED]*$\times 2$ *[13].*

A degenerate sharpness witness is also formalized: `branch2_witness_s49` gives a genuine branch-2 identity at $s = 49$, $m = 7$, $A = r_1 = 1$ ($p = 145860$, $q = 4$) — branch 2 is possible but parametrically starved, as the theorem predicts.

**Refuted claim** (R209 (4), the mod-8 conclusion; REFUTED)**.** The claim "$m \le 2\gcd(\text{XSTEP}, (s+m)/4)$, hence $m \equiv 7 \pmod 8$" is false as stated. Its proof argues "$m \ge 3$ forces $g \ge 2$, so $2 \mid x_0$" — but $g \ge 2$ does not imply $2 \mid g$. The formalized counterexample (`branch2_m_mod8_refuted`): XSTEP $= 15$, $s = 97$, $m = 3$, $A = 10$, $r_1 = 2$, $p = 3$, $q = 5$, $x_0 = 25$, $M = 60$ — every hypothesis of (4) holds by `decide` (here $g = \gcd(15, 25) = 5 \ge 2$ but $2 \nmid 25 = x_0$, the exact gap), the branch-2 identity holds for all $t$ (cleared numerator identically 0, independently checked in sympy; at $t = 0$: $4/97 = 1/25 + 1/4850 + 1/970$), yet $m = 3 \not\equiv 7 \pmod 8$. The bound $m \le 2g$ itself stands; the mod-8 inference is withdrawn, and any argument citing it inherits the gap [14].

## 5 Discussion

**What the five pieces share.** Fiber-compatibility is the local inverse-closure law; the composite lift shows it composes through CRT only when the $D \mid n$ hypothesis holds; the Correlation Break shows that without joint digit data, component-wise compatibility is worthless — a lesson with the flavor of a failure of local-to-global for inverse-closure. Suppression theory then studies the *dynamics* of the resulting correlation holes under refinement: $\kappa$ measures hole survival, $\Delta$-gating characterizes which destruction mechanisms a step even allows, and T8 draws the line where prediction stops (fine-level data is genuinely fine). The tiling catalogue closes the last two fibers of an independent covering line, and the branch-2 work collapses a supposedly cubic family to linear algebra plus integrality, then exhibits one infinite parametric family while honestly retracting one over-claim.

**Significance.** (1) The fiber-compatibility iff is the sharp characterization the program lacked: saturation sufficient, asymmetry a counting refutation, $m = 8$ the witness that unsaturated need not break — and it is now the kernel everything downstream (composites, suppression) builds on, in machine-checked form. (2) The correlation-hole phenomenon is new structural content: 12/44 real moduli and minimal 2-element witnesses show the naive CRT reduction fails systematically, not pathologically. (3) T8's impossibility is a genuine boundary result for the program: no coarse-feature function predicts $\kappa$; the $225 \to 675$ vs $1125 \to 3375$ witness pair agrees on all six coarse features and differs only in fine digit-set saturation. (4) The branch-2 collapse reframes an entire search lane: R67/R71's zero branch-2 hits were a search-box artifact ($r < M$ provably vacuous), and the surviving content is integrality of two parameters. (5) The mod-8 refutation is the honesty discipline working as designed: machine-checking caught a gap ($g \ge 2 \not\Rightarrow 2 \mid g$) that hand review missed, the [PROVED*] label was withdrawn, and the corrected record stands.

**Limitations.** (i) The full $\pi(P)$ inverse-closed $\iff$ compat theorem at the dossier level is [PROVED] in its kernel and lift-set packaging; the R192 hand proof of the composite iff and the T2(b)/T3/T5–T8 mechanism proofs remain [PROVED*] with [COMPUTED] empirical content. (ii) The no-tiling theorem is conditional on the cited [PROVED*] premises and the [COMPUTED] verdict table (Assumptions 2, 3). (iii) The 10 OPEN micro-instances are individually undecided; the $13 \mid h$ regime, un-thinned quads, and the full ES conjecture remain [OPEN]. (iv) A corrected parity/shape restriction on the branch-2 parameter $m$ (replacing the refuted (4)) is open; the 196 refinements unreachable at $M = 2042040$ need a deeper modulus or a new splitting form. (v) Whether $\Delta$-gating admits a sufficient condition for storms (R207 §6) and whether fine starvation is predictable (the starvation law) are open mathematical targets, not formalization ones.

## 6 Conclusion

We have presented five machine-checked components of the Erdős–Straus covering program: the fiber-compatibility kernel and its lift-set packaging (`ESFiberCompat.lean`); the composite lift with the $D \mid n$ lemma and the Correlation Break (`ESFiberCompatComposite.lean`); the suppression coefficient, persistence rate, demands/capability kernels, and the R-operator section (`CompositeSuppression.lean`); the fiber-$\{6, 11\}$ catalogue and the uniform no-tiling theorem (`ESFiberTiling.lean`); and the branch-2 degeneracy theorems, the divisor-closed family identity, and the formal refutation of the mod-8 claim (`ESBranchTwo.lean`) — all in Lean 4.34.1 + Mathlib v4.34.1, `lake build` exit 0 on bakkie, zero sorrys, standard-only axioms. The deductive core is fully machine-checked; the computational strata (empirical iffs, census aggregates, SAT verdicts, the 92-identity battery) are labeled [COMPUTED] and kept separate from the proofs; one stated conjecture is reported refuted. The natural next steps are the [PROVED*]→[PROVED] promotion of the composite-iff hand proof and the suppression mechanism proofs, a corrected branch-2 parity law, and the 196 unreachable refinements at a deeper modulus.

## 7 References

## References


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